

数学竞赛系列分类题库集 V1.0

AMC 10 分类题库

一、平面几何

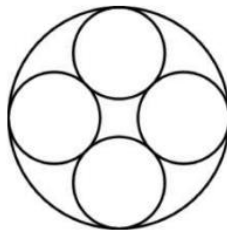
Problem 17

Rectangle $ABCD$ has $AB = 4$ and $BC = 3$. Segment EF is constructed through B so that EF is perpendicular to DB , and A and C lie on DE and DF , respectively. What is EF ?

- (A) 9 (B) 10 (C) $\frac{125}{12}$ (D) $\frac{103}{9}$ (E) 12

Problem 21

Many Gothic cathedrals have windows with portions containing a ring of congruent circles that are circumscribed by a larger circle. In the figure shown, the number of smaller circles is four. What is the ratio of the sum of the areas of the four smaller circles to the area of the larger circle?



- (A) $3 - 2\sqrt{2}$ (B) $2 - \sqrt{2}$ (C) $4(3 - 2\sqrt{2})$ (D) $\frac{1}{2}(3 - \sqrt{2})$ (E) $2\sqrt{2} - 2$

Problem 23

Convex quadrilateral $ABCD$ has $AB = 9$ and $CD = 12$. Diagonals AC and BD intersect at E , $AC = 14$, and $\triangle AED$ and $\triangle BEC$ have equal areas. What is AE ?

- (A) $\frac{9}{2}$ (B) $\frac{50}{11}$ (C) $\frac{21}{4}$ (D) $\frac{17}{3}$ (E) 6

09b

Problem 16

Points A and C lie on a circle centered at O , each of $\overline{BA}$ and $\overline{BC}$ are tangent to the circle, and $\triangle ABC$ is equilateral. The circle intersects $\overline{BO}$ at D . What is $\frac{BD}{BO}$?

- (A) $\frac{\sqrt{2}}{3}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{3}}{3}$ (D) $\frac{\sqrt{2}}{2}$ (E) $\frac{\sqrt{3}}{2}$

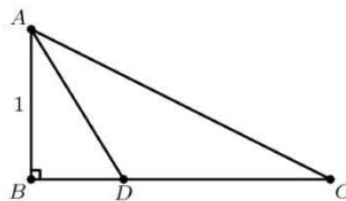
Problem 18

Rectangle $ABCD$ has $AB = 8$ and $BC = 6$. Point M is the midpoint of diagonal $\overline{AC}$, and E is on AB with $\overline{ME} \perp \overline{AC}$. What is the area of $\triangle AME$?

- (A) $\frac{65}{8}$ (B) $\frac{25}{3}$ (C) 9 (D) $\frac{75}{8}$ (E) $\frac{85}{8}$

Problem 20

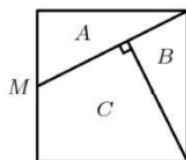
Triangle ABC has a right angle at B , $AB = 1$, and $BC = 2$. The bisector of $\angle BAC$ meets $\overline{BC}$ at D . What is BD ?



- (A) $\frac{\sqrt{3}-1}{2}$ (B) $\frac{\sqrt{5}-1}{2}$ (C) $\frac{\sqrt{5}+1}{2}$ (D) $\frac{\sqrt{6}+\sqrt{2}}{2}$ (E) $2\sqrt{3}-1$

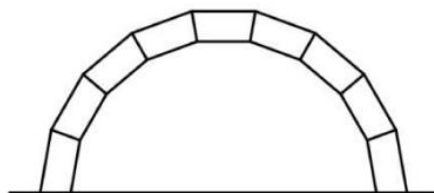
Problem 22

A cubical cake with edge length 2 inches is iced on the sides and the top. It is cut vertically into three pieces as shown in this top view, where M is the midpoint of a top edge. The piece whose top is triangle B contains c cubic inches of cake and s square inches of icing. What is $c + s$?



- (A) $\frac{24}{5}$ (B) $\frac{32}{5}$ (C) $8 + \sqrt{5}$ (D) $5 + \frac{16\sqrt{5}}{5}$ (E) $10 + 5\sqrt{5}$

The keystone arch is an ancient architectural feature. It is composed of congruent isosceles trapezoids fitted together along the non-parallel sides, as shown. The bottom sides of the two end trapezoids are horizontal. In an arch made with 9 trapezoids, let x be the angle measure in degrees of the larger interior angle of the trapezoid. What is x ?



- (A) 100 (B) 102 (C) 104 (D) 106 (E) 108

10a

Problem 16

Nondegenerate $\triangle ABC$ has integer side lengths, $\overline{BD}$ is an angle bisector, $AD = 3$, and $DC = 8$. What is the smallest possible value of the perimeter?

- (A) 30 (B) 33 (C) 35 (D) 36 (E) 37

Problem 19

Equiangular hexagon $ABCDEF$ has side lengths $AB = CD = EF = 1$ and $BC = DE = FA = r$. The area of $\triangle ACE$ is 70% of the area of the hexagon. What is the sum of all possible values of r ?

- (A) $\frac{4\sqrt{3}}{3}$ (B) $\frac{10}{3}$ (C) 4 (D) $\frac{17}{4}$ (E) 6

10b

Problem 16

A square of side length 1 and a circle of radius $\frac{\sqrt{3}}{3}$ share the same center. What is the area inside the circle, but outside the square?

- (A) $\frac{\pi}{3} - 1$ (B) $\frac{2\pi}{9} - \frac{\sqrt{3}}{3}$ (C) $\frac{\pi}{18}$ (D) $\frac{1}{4}$ (E) $\frac{2\pi}{9}$

Problem 19

A circle with center O has area 156π . Triangle ABC is equilateral, $\overline{BC}$ is a chord on the circle, $OA = 4\sqrt{3}$, and point O is outside $\triangle ABC$. What is the side length of $\triangle ABC$?

- (A) $2\sqrt{3}$ (B) 6 (C) $4\sqrt{3}$ (D) 12 (E) 18

Problem 20

Two circles lie outside regular hexagon $ABCDEF$. The first is tangent to $\overline{AB}$, and the second is tangent to $\overline{DE}$. Both are tangent to lines BC and FA . What is the ratio of the area of the second circle to that of the first circle?

- (A) 18 (B) 27 (C) 36 (D) 81 (E) 108

11a

Problem 18

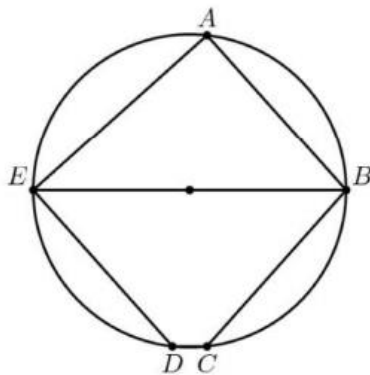
Circles A , B , and C each have radius 1 . Circles A and B share one point of tangency. Circle C has a point of tangency with the midpoint of $\overline{AB}$. What is the area inside Circle C but outside circle A and circle B ?

- (A) $3 - \frac{\pi}{2}$ (B) $\frac{\pi}{2}$ (C) 2 (D) $\frac{3\pi}{4}$ (E) $1 + \frac{\pi}{2}$

11b

Problem 17

In the given circle, the diameter $\overline{EB}$ is parallel to $\overline{DC}$, and $\overline{AB}$ is parallel to $\overline{ED}$. The angles AEB and ABE are in the ratio $4:5$. What is the degree measure of angle BCD ?



- (A) 120 (B) 125 (C) 130 (D) 135 (E) 140

Problem 18

Rectangle $ABCD$ has $AB = 6$ and $BC = 3$. Point M is chosen on side AB so that $\angle AMD = \angle CMD$. What is the degree measure of $\angle AMD$?

- (A) 15 (B) 30 (C) 45 (D) 60 (E) 75

Problem 20

Rhombus $ABCD$ has side length 2 and $\angle B = 120^\circ$. Region R consists of all points inside the rhombus that are closer to vertex B than any of the other three vertices. What is the area of R ?

- (A) $\frac{\sqrt{3}}{3}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{2\sqrt{3}}{3}$ (D) $1 + \frac{\sqrt{3}}{3}$ (E) 2

Problem 25

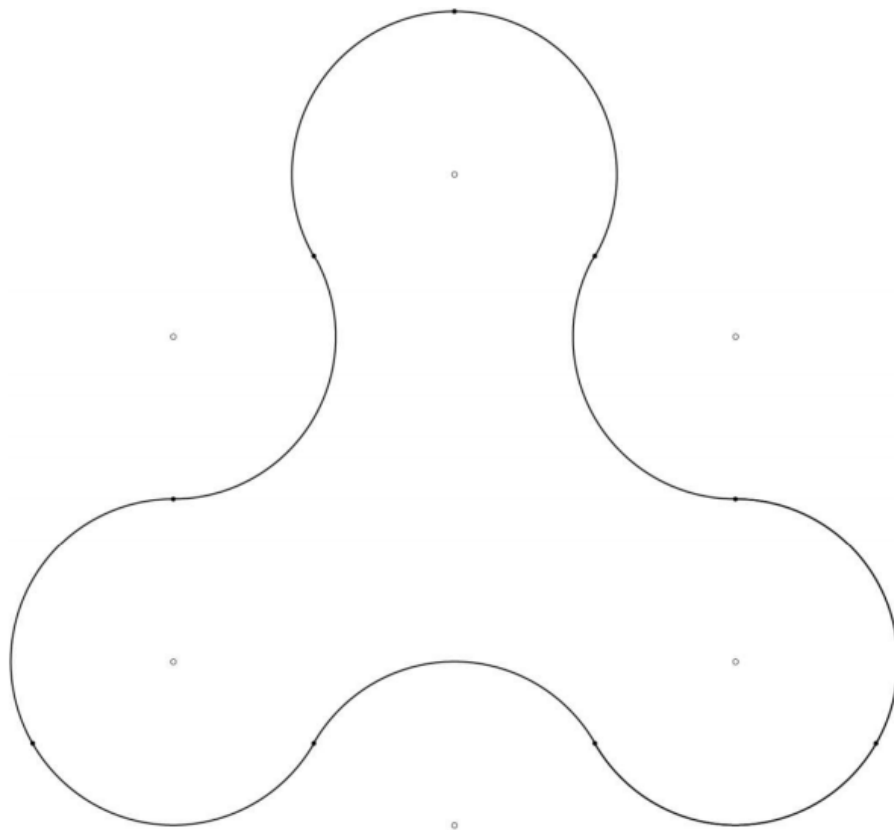
Let T_1 be a triangle with sides 2011 , 2012 , and 2013 . For $n \geq 1$, if $T_n = \triangle ABC$ and D , E , and F are the points of tangency of the incircle of $\triangle ABC$ to the sides AB , BC and AC , respectively, then T_{n+1} is a triangle with side lengths AD , BE , and CF , if it exists. What is the perimeter of the last triangle in the sequence (T_n) ?

- (A) $\frac{1509}{8}$ (B) $\frac{1509}{32}$ (C) $\frac{1509}{64}$ (D) $\frac{1509}{128}$ (E) $\frac{1509}{256}$

12a

Problem 18

The closed curve in the figure is made up of 9 congruent circular arcs each of length $\frac{2\pi}{3}$, where each of the centers of the corresponding circles is among the vertices of a regular hexagon of side 2 . What is the area enclosed by the curve?



- (A) $2\pi + 6$ (B) $2\pi + 4\sqrt{3}$ (C) $3\pi + 4$ (D) $2\pi + 3\sqrt{3} + 2$ (E) $\pi + 6\sqrt{3}$

Problem 21

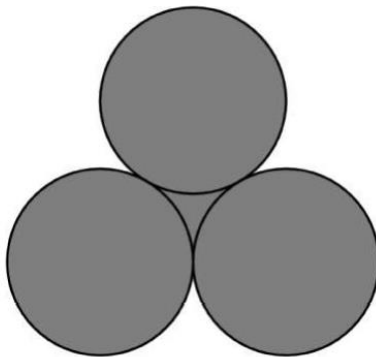
Let points $A = (0, 0, 0)$, $B = (1, 0, 0)$, $C = (0, 2, 0)$, and $D = (0, 0, 3)$. Points E , F , G , and H are midpoints of line segments $\overline{BD}$, $\overline{AB}$, $\overline{AC}$, and $\overline{DC}$ respectively. What is the area of $EFGH$?

- (A) $\sqrt{2}$ (B) $\frac{2\sqrt{5}}{3}$ (C) $\frac{3\sqrt{5}}{4}$ (D) $\sqrt{3}$ (E) $\frac{2\sqrt{7}}{3}$

12b

Problem 16

Three circles with radius 2 are mutually tangent. What is the total area of the circles and the region bounded by them, as shown in the figure?



- (A) $10\pi + 4\sqrt{3}$ (B) $13\pi - \sqrt{3}$ (C) $12\pi + \sqrt{3}$ (D) $10\pi + 9$ (E) 13π

Problem 17

Jesse cuts a circular paper disk of radius 12 along two radii to form two sectors, the smaller having a central angle of 120 degrees. He makes two circular cones, using each sector to form the lateral surface of a cone. What is the ratio of the volume of the smaller cone to that of the larger?

- (A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{\sqrt{10}}{10}$ (D) $\frac{\sqrt{5}}{6}$ (E) $\frac{\sqrt{5}}{5}$

Problem 18

Suppose that one of every 500 people in a certain population has a particular disease, which displays no symptoms. A blood test is available for screening for this disease. For a person who has this disease, the test always turns out positive. For a person who does not have the disease, however, there is a 2% false positive rate—in other words, for such people, 98% of the time the test will turn out negative, but 2% of the time the test will turn out positive and will incorrectly indicate that the person has the disease. Let p be the probability that a person who is chosen at random from this population and gets a positive test result actually has the disease. Which of the following is closest to p ?

- (A) $\frac{1}{98}$ (B) $\frac{1}{9}$ (C) $\frac{1}{11}$ (D) $\frac{49}{99}$ (E) $\frac{98}{99}$

Problem 19

In rectangle $ABCD$, $AB = 6$, $AD = 30$, and G is the midpoint of $\overline{AD}$. Segment $\overline{AB}$ is extended 2 units beyond B to point E , and F is the intersection of $\overline{ED}$ and $\overline{BC}$. What is the area of $BFDG$?

- (A) $\frac{133}{2}$ (B) 67 (C) $\frac{135}{2}$ (D) 68 (E) $\frac{137}{2}$

Problem 21

Four distinct points are arranged on a plane so that the segments connecting them have lengths a , a , a , a , $2a$, and b . What is the ratio of b to a ?

- (A) $\sqrt{3}$ (B) 2 (C) $\sqrt{5}$ (D) 3 (E) π

13a

Problem 16

A triangle with vertices $(6, 5)$, $(8, -3)$, and $(9, 1)$ is reflected about the line $x = 8$ to create a second triangle. What is the area of the union of the two triangles?

- (A) 9 (B) $\frac{28}{3}$ (C) 10 (D) $\frac{31}{3}$ (E) $\frac{32}{3}$

Problem 20

A unit square is rotated 45° about its center. What is the area of the region swept out by the interior of the square?

- (A) $1 - \frac{\sqrt{2}}{2} + \frac{\pi}{4}$ (B) $\frac{1}{2} + \frac{\pi}{4}$ (C) $2 - \sqrt{2} + \frac{\pi}{4}$
 (D) $\frac{\sqrt{2}}{2} + \frac{\pi}{4}$ (E) $1 + \frac{\sqrt{2}}{4} + \frac{\pi}{8}$

Problem 23

In $\triangle ABC$, $\overline{AB} = 86$, and $\overline{AC} = 97$. A circle with center A and radius AB intersects $\overline{BC}$ at points B and X . Moreover $\overline{BX}$ and $\overline{CX}$ have integer lengths. What is BC ?

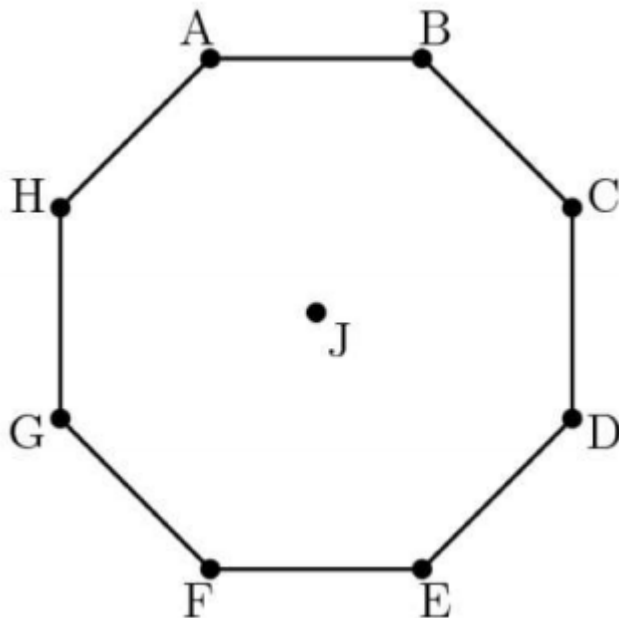
- (A) 11 (B) 28 (C) 33 (D) 61 (E) 72

13b

Problem 16

In triangle ABC , medians AD and CE intersect at P , $PE = 1.5$, $PD = 2$ and $DE = 2.5$. What is the area of $AEDC$?

- (A) 13 (B) 13.5 (C) 14 (D) 14.5 (E) 15



Problem 23

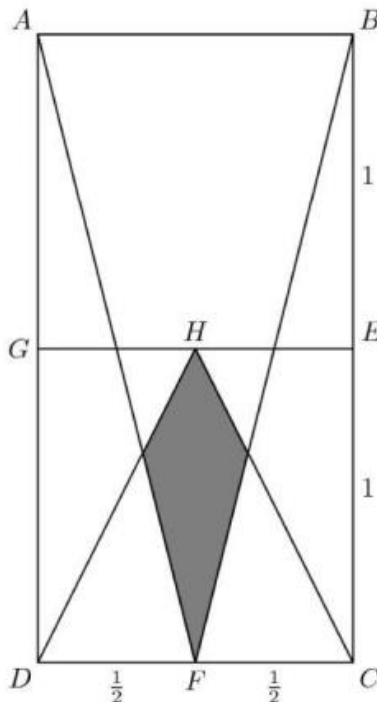
In triangle ABC , $AB = 13$, $BC = 14$, and $CA = 15$. Distinct points D , E , and F lie on segments $\overline{BC}$, $\overline{CA}$, and $\overline{DE}$, respectively, such that $\overline{AD} \perp \overline{BC}$, $\overline{DE} \perp \overline{AC}$, and $\overline{AF} \perp \overline{BF}$. The length of segment $\overline{DF}$ can be written as $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?

- (A) 18 (B) 21 (C) 24 (D) 27 (E) 30

14a

Problem 16

In rectangle $ABCD$, $AB = 1$, $BC = 2$, and points E , F , and G are midpoints of $\overline{BC}$, $\overline{CD}$, and $\overline{AD}$, respectively. Point H is the midpoint of $\overline{GE}$. What is the area of the shaded region?



- (A) $\frac{1}{12}$ (B) $\frac{\sqrt{3}}{18}$ (C) $\frac{\sqrt{2}}{12}$ (D) $\frac{\sqrt{3}}{12}$ (E) $\frac{1}{6}$

Problem 18

A square in the coordinate plane has vertices whose y -coordinates are 0, 1, 4, and 5. What is the area of the square?

- (A) 16 (B) 17 (C) 25 (D) 26 (E) 27

Problem 22

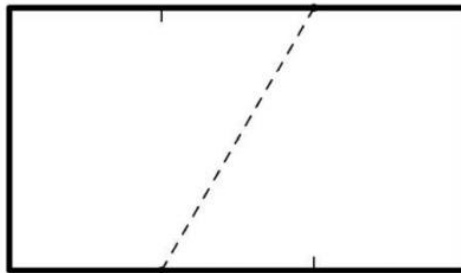
In rectangle $ABCD$, $AB = 20$ and $BC = 10$. Let E be a point on $\overline{CD}$ such that $\angle CBE = 15^\circ$. What is AE ?

- (A) $\frac{20\sqrt{3}}{3}$ (B) $10\sqrt{3}$ (C) 18 (D) $11\sqrt{3}$ (E) 20

Problem 23

A rectangular piece of paper whose length is $\sqrt{3}$ times the width has area A . The paper is divided into three equal sections along the opposite lengths, and then a dotted line is drawn from the first divider to the second divider on the opposite side as shown. The paper is then folded flat along this dotted line to create a new shape

with area B . What is the ratio $B : A$?



- (A) 1 : 2 (B) 3 : 5 (C) 2 : 3 (D) 3 : 4 (E) 4 : 5

14b

Problem 21

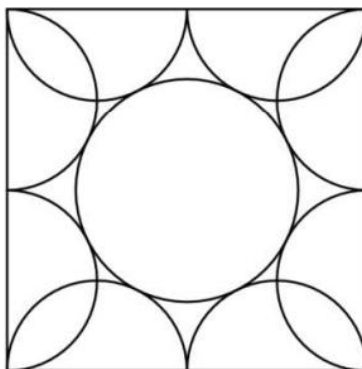
Trapezoid $ABCD$ has parallel sides $\overline{AB}$ of length 33 and $\overline{CD}$ of length 21. The other two sides are of lengths 10 and 14. The angles at A and B are acute. What is the length of the shorter diagonal of $ABCD$?

- (A) $10\sqrt{6}$ (B) 25 (C) $8\sqrt{10}$ (D) $18\sqrt{2}$ (E) 26

Problem 22

Eight semicircles line the inside of a square with side length 2 as shown. What is the radius of the circle tangent to all of these semicircles?

- (A) $\frac{1+\sqrt{2}}{4}$ (B) $\frac{\sqrt{5}-1}{2}$ (C) $\frac{\sqrt{3}+1}{4}$ (D) $\frac{2\sqrt{3}}{5}$ (E) $\frac{\sqrt{5}}{3}$



15a

Problem 19

The isosceles right triangle ABC has right angle at C and area 12.5. The rays trisecting $\angle ACB$ intersect AB at D and E . What is the area of $\triangle CDE$?

- (A) $\frac{5\sqrt{2}}{3}$ (B) $\frac{50\sqrt{3}-75}{4}$ (C) $\frac{15\sqrt{3}}{8}$ (D) $\frac{50-25\sqrt{3}}{2}$ (E) $\frac{25}{6}$

Problem 25

Let S be a square of side length 1. Two points are chosen independently at random on the sides of S . The probability that the straight-line distance between the points is at least $\frac{1}{2}$ is $\frac{a-b\pi}{c}$, where a , b , and c are positive integers with $\gcd(a, b, c) = 1$. What is $a + b + c$?

- (A) 59 (B) 60 (C) 61 (D) 62 (E) 63

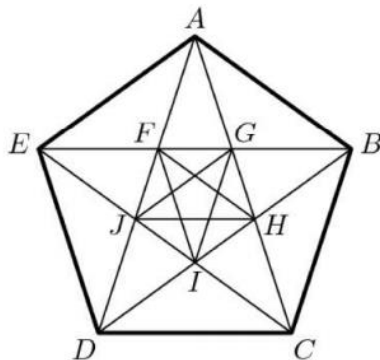
Problem 19

In $\triangle ABC$, $\angle C = 90^\circ$ and $AB = 12$. Squares $ABXY$ and $ACWZ$ are constructed outside of the triangle. The points X, Y, Z , and W lie on a circle. What is the perimeter of the triangle?

- (A) $12 + 9\sqrt{3}$ (B) $18 + 6\sqrt{3}$ (C) $12 + 12\sqrt{2}$ (D) 30 (E) 32

Problem 22

In the figure shown below, $ABCDE$ is a regular pentagon and $AG = 1$. What is $FG + JH + CD$?



- (A) 3 (B) $12 - 4\sqrt{5}$ (C) $\frac{5 + 2\sqrt{5}}{3}$ (D) $1 + \sqrt{5}$ (E) $\frac{11 + 11\sqrt{5}}{10}$

16a

Problem 16

A triangle with vertices $A(0, 2)$, $B(-3, 2)$, and $C(-3, 0)$ is reflected about the x -axis, then the image $\triangle A'B'C'$ is rotated counterclockwise about the origin by 90° to produce $\triangle A''B''C''$. Which of the following transformations will return $\triangle A''B''C''$ to $\triangle ABC$?

- (A) counterclockwise rotation about the origin by 90° .
 (B) clockwise rotation about the origin by 90° .
 (C) reflection about the x -axis
 (D) reflection about the line $y = x$
 (E) reflection about the y -axis.

Problem 19

In rectangle $ABCD$, $AB = 6$ and $BC = 3$. Point E between B and C , and point F between E and C are such that $BE = EF = FC$. Segments $\overline{AE}$ and $\overline{AF}$ intersect $\overline{BD}$ at P and Q , respectively. The ratio $BP : PQ : QD$ can be written as $r : s : t$ where the greatest common factor of r , s and t is 1. What is $r + s + t$?

- (A) 7 (B) 9 (C) 12 (D) 15 (E) 20

Problem 21

Circles with centers P , Q and R , having radii 1, 2 and 3, respectively, lie on the same side of line l and are tangent to l at P' , Q' and R' , respectively, with Q' between P' and R' . The circle with center Q is externally tangent to each of the other two circles. What is the area of triangle PQR ?

- (A) 0 (B) $\sqrt{\frac{2}{3}}$ (C) 1 (D) $\sqrt{6} - \sqrt{2}$ (E) $\sqrt{\frac{3}{2}}$

Problem 24

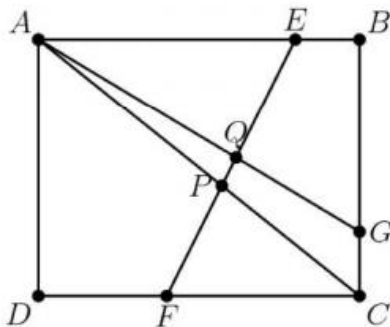
A quadrilateral is inscribed in a circle of radius $200\sqrt{2}$. Three of the sides of this quadrilateral have length 200. What is the length of the fourth side?

- (A) 200 (B) $200\sqrt{2}$ (C) $200\sqrt{3}$ (D) $300\sqrt{2}$ (E) 500

16b

Problem 19

Rectangle $ABCD$ has $AB = 5$ and $BC = 4$. Point E lies on $\overline{AB}$ so that $\frac{EB}{AB} = \frac{1}{5}$, point G lies on $\overline{BC}$ so that $\frac{CG}{BC} = \frac{1}{4}$, and point F lies on $\overline{CD}$ so that $\frac{DF}{CD} = \frac{2}{5}$. Segments $\overline{AG}$ and $\overline{AC}$ intersect $\overline{EF}$ at Q and P , respectively. What is the value of $\frac{PQ}{EF}$?



- (A) $\frac{\sqrt{13}}{16}$ (B) $\frac{\sqrt{2}}{13}$ (C) $\frac{9}{82}$ (D) $\frac{10}{91}$ (E) $\frac{1}{9}$

Problem 23

In regular hexagon $ABCDEF$, points W , X , Y , and Z are chosen on sides $\overline{BC}$, $\overline{CD}$, $\overline{EF}$, and $\overline{FA}$ respectively, so lines $\overline{AB}$, $\overline{ZW}$, $\overline{YX}$, and $\overline{ED}$ are parallel and equally spaced. What is the ratio of the area of hexagon $WCXYFZ$ to the area of hexagon $ABCDEF$?

- (A) $\frac{1}{3}$ (B) $\frac{10}{27}$ (C) $\frac{11}{27}$ (D) $\frac{4}{9}$ (E) $\frac{13}{27}$

17a

Problem 21

A square with side length x is inscribed in a right triangle with sides of length 3, 4, and 5 so that one vertex of the square coincides with the right-angle vertex of the triangle. A square with side length y is inscribed in another right triangle with sides of length 3, 4, and 5 so that one side of the square lies on the hypotenuse of the triangle. What is $\frac{x}{y}$?

- (A) $\frac{12}{13}$ (B) $\frac{35}{37}$ (C) 1 (D) $\frac{37}{35}$ (E) $\frac{13}{12}$

Problem 22

Sides $\overline{AB}$ and $\overline{AC}$ of equilateral triangle ABC are tangent to a circle at points B and C respectively. What fraction of the area of $\triangle ABC$ lies outside the circle?

- (A) $\frac{4\sqrt{3}\pi}{27} - \frac{1}{3}$ (B) $\frac{\sqrt{3}}{2} - \frac{\pi}{8}$ (C) $\frac{1}{2}$ (D) $\sqrt{3} - \frac{2\sqrt{3}\pi}{9}$ (E) $\frac{4}{3} - \frac{4\sqrt{3}\pi}{27}$

17b

Problem 19

Let ABC be an equilateral triangle. Extend side $\overline{AB}$ beyond B to a point B' so that $BB' = 3AB$. Similarly, extend side $\overline{BC}$ beyond C to a point C' so that $CC' = 3BC$, and extend side $\overline{CA}$ beyond A to a point A' so that $AA' = 3CA$. What is the ratio of the area of $\triangle A'B'C'$ to the area of $\triangle ABC$?

- (A) 9 : 1 (B) 16 : 1 (C) 25 : 1 (D) 36 : 1 (E) 37 : 1

Problem 21

In $\triangle ABC$, $AB = 6$, $AC = 8$, $BC = 10$, and D is the midpoint of $\overline{BC}$. What is the sum of the radii of the circles inscribed in $\triangle ADB$ and $\triangle ADC$?

- (A) $\sqrt{5}$ (B) $\frac{11}{4}$ (C) $2\sqrt{2}$ (D) $\frac{17}{6}$ (E) 3

Problem 22

The diameter AB of a circle of radius 2 is extended to a point D outside the circle so that $BD = 3$. Point E is chosen so that $ED = 5$ and line ED is perpendicular to line AD . Segment AE intersects the circle at a point C between A and E . What is the area of $\triangle ABC$?

- (A) $\frac{120}{37}$ (B) $\frac{140}{39}$ (C) $\frac{145}{39}$ (D) $\frac{140}{37}$ (E) $\frac{120}{31}$

Problem 24

The vertices of an equilateral triangle lie on the hyperbola $xy = 1$, and a vertex of this hyperbola is the centroid of the triangle. What is the square of the area of the triangle?

- (A) 48 (B) 60 (C) 108 (D) 120 (E) 169

18a

Problem 16

Right triangle ABC has leg lengths $AB = 20$ and $BC = 21$. Including $\overline{AB}$ and $\overline{BC}$, how many line segments with integer length can be drawn from vertex B to a point on hypotenuse $\overline{AC}$?

- (A) 5 (B) 8 (C) 12 (D) 13 (E) 15

Problem 20

A scanning code consists of a 7×7 grid of squares, with some of its squares colored black and the rest colored white. There must be at least one square of each color in this grid of 49 squares. A scanning code is called *symmetric* if its look does not change when the entire square is rotated by a multiple of 90° counterclockwise around its center, nor when it is reflected across a line joining opposite corners or a line joining midpoints of opposite sides. What is the total number of possible symmetric scanning codes?

- (A) 510 (B) 1022 (C) 8190 (D) 8192 (E) 65,534

Problem 24

Triangle ABC with $AB = 50$ and $AC = 10$ has area 120. Let D be the midpoint of $\overline{AB}$, and let E be the midpoint of $\overline{AC}$. The angle bisector of $\angle BAC$ intersects $\overline{DE}$ and $\overline{BC}$ at F and G , respectively. What is the area of quadrilateral $FDBG$?

- (A) 60 (B) 65 (C) 70 (D) 75 (E) 80

18b

Problem 17

In rectangle $PQRS$, $PQ = 8$ and $QR = 6$. Points A and B lie on $\overline{PQ}$, points C and D lie on $\overline{QR}$, points E and F lie on $\overline{RS}$, and points G and H lie on $\overline{SP}$ so that $AP = BQ < 4$ and the convex octagon $ABCDEFGH$ is equilateral. The length of a side of this octagon can be expressed in the form $k + m\sqrt{n}$, where k , m , and n are integers and n is not divisible by the square of any prime. What is $k + m + n$?

- (A) 1 (B) 7 (C) 21 (D) 92 (E) 106

Problem 24

Let $ABCDEF$ be a regular hexagon with side length 1. Denote by X , Y , and Z the midpoints of sides $\overline{AB}$, $\overline{CD}$, and $\overline{EF}$, respectively. What is the area of the convex hexagon whose interior is the intersection of the interiors of $\triangle ACE$ and $\triangle XYZ$?

- (A) $\frac{3}{8}\sqrt{3}$ (B) $\frac{7}{16}\sqrt{3}$ (C) $\frac{15}{32}\sqrt{3}$ (D) $\frac{1}{2}\sqrt{3}$ (E) $\frac{9}{16}\sqrt{3}$

二、立体几何

09b

10a

Problem 17

A solid cube has side length 3 inches. A 2-inch by 2-inch square hole is cut into the center of each face. The edges of each cut are parallel to the edges of the cube, and each hole goes all the way through the cube. What is the volume, in cubic inches, of the remaining solid?

- (A) 7 (B) 8 (C) 10 (D) 12 (E) 15

Problem 20

A fly trapped inside a cubical box with side length 1 meter decides to relieve its boredom by visiting each corner of the box. It will begin and end in the same corner and visit each of the other corners exactly once. To get from a corner to any other corner, it will either fly or crawl in a straight line. What is the maximum possible length, in meters, of its path?

- (A) $4 + 4\sqrt{2}$ (B) $2 + 4\sqrt{2} + 2\sqrt{3}$ (C) $2 + 3\sqrt{2} + 3\sqrt{3}$
(D) $4\sqrt{2} + 4\sqrt{3}$ (E) $3\sqrt{2} + 5\sqrt{3}$

10b

11a

Problem 24

Two distinct regular tetrahedra have all their vertices among the vertices of the same unit cube. What is the volume of the region formed by the intersection of the tetrahedra?

- (A) $\frac{1}{12}$ (B) $\frac{\sqrt{2}}{12}$ (C) $\frac{\sqrt{3}}{12}$ (D) $\frac{1}{6}$ (E) $\frac{\sqrt{2}}{6}$

11b

Problem 22

A pyramid has a square base with sides of length 1 and has lateral faces that are equilateral triangles. A cube is placed within the pyramid so that one face is on the base of the pyramid and its opposite face has all its edges on the lateral faces of the pyramid. What is the volume of this cube?

- (A) $5\sqrt{2} - 7$ (B) $7 - 4\sqrt{3}$ (C) $\frac{2\sqrt{2}}{27}$ (D) $\frac{\sqrt{2}}{9}$ (E) $\frac{\sqrt{3}}{9}$

12a

12b

Problem 23

A solid tetrahedron is sliced off a wooden unit cube by a plane passing through two nonadjacent vertices on one face and one vertex on the opposite face not adjacent to either of the first two vertices. The tetrahedron is discarded and the remaining portion of the cube is placed on a table with the cut surface face down. What is the height of this object?

- (A) $\frac{\sqrt{3}}{3}$ (B) $\frac{2\sqrt{2}}{3}$ (C) 1 (D) $\frac{2\sqrt{3}}{3}$ (E) $\sqrt{2}$

13a

Problem 22

Six spheres of radius **1** are positioned so that their centers are at the vertices of a regular hexagon of side length **2**. The six spheres are internally tangent to a larger sphere whose center is the center of the hexagon. An eighth sphere is externally tangent to the six smaller spheres and internally tangent to the larger sphere. What is the radius of this eighth sphere?

- (A) $\sqrt{2}$ (B) $\frac{3}{2}$ (C) $\frac{5}{3}$ (D) $\sqrt{3}$ (E) 2

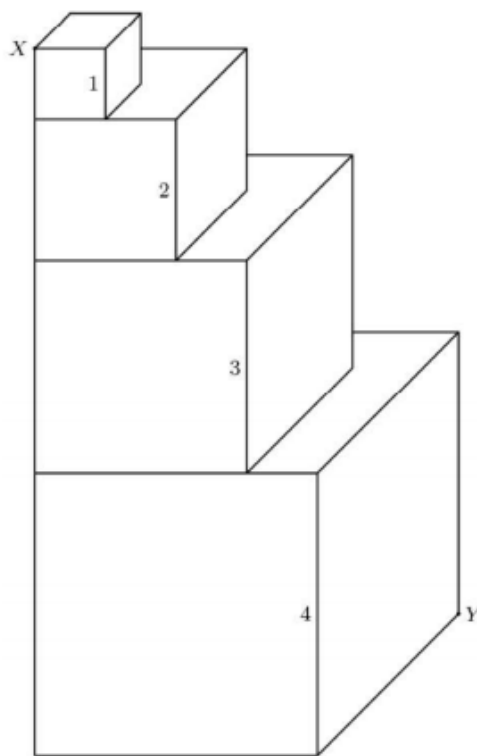
13b

14a

Problem 19

Four cubes with edge lengths **1**, **2**, **3**, and **4** are stacked as shown. What is the length of the portion of $\overline{XY}$ contained in the cube with edge length **3**?

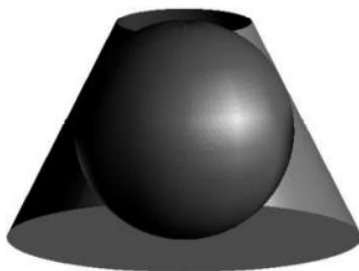
- (A) $\frac{3\sqrt{33}}{5}$ (B) $2\sqrt{3}$ (C) $\frac{2\sqrt{33}}{3}$ (D) 4 (E) $3\sqrt{2}$



14b

Problem 23

A sphere is inscribed in a truncated right circular cone as shown. The volume of the truncated cone is twice that of the sphere. What is the ratio of the radius of the bottom base of the truncated cone to the radius of the top base of the truncated cone?



- (A) $\frac{3}{2}$ (B) $\frac{1+\sqrt{5}}{2}$ (C) $\sqrt{3}$ (D) 2 (E) $\frac{3+\sqrt{5}}{2}$

15a

Problem 21

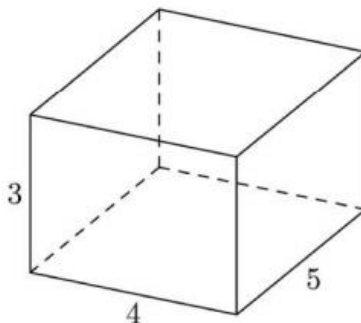
Tetrahedron $ABCD$ has $AB = 5$, $AC = 3$, $BC = 4$, $BD = 4$, $AD = 3$, and $CD = \frac{12}{5}\sqrt{2}$. What is the volume of the tetrahedron?

- (A) $3\sqrt{2}$ (B) $2\sqrt{5}$ (C) $\frac{24}{5}$ (D) $3\sqrt{3}$ (E) $\frac{24}{5}\sqrt{2}$

15b

Problem 17

The centers of the faces of the right rectangular prism shown below are joined to create an octahedron. What is the volume of the octahedron?



- (A) $\frac{75}{12}$ (B) 10 (C) 12 (D) $10\sqrt{2}$ (E) 15

Problem 20

Erin the ant starts at a given corner of a cube and crawls along exactly 7 edges in such a way that she visits every corner exactly once and then finds that she is unable to return along an edge to her starting point. How many paths are there meeting these conditions?

- (A) 6 (B) 9 (C) 12 (D) 18 (E) 24

16a

Problem 18

Each vertex of a cube is to be labeled with an integer 1 through 8, with each integer being used once, in such a way that the sum of the four numbers on the vertices of a face is the same for each face. Arrangements that can be obtained from each other through rotations of the cube are considered to be the same. How many different arrangements are possible?

- (A) 1 (B) 3 (C) 6 (D) 12 (E) 24

三、代数

Problem 16

Let a , b , c , and d be real numbers with $|a - b| = 2$, $|b - c| = 3$, and $|c - d| = 4$. What is the sum of all possible values of $|a - d|$?

- (A) 9 (B) 12 (C) 15 (D) 18 (E) 24

Problem 25

For $k > 0$, let $I_k = 10 \dots 064$, where there are k zeros between the 1 and the 6. Let $N(k)$ be the number of factors of 2 in the prime factorization of I_k . What is the maximum value of $N(k)$?

- (A) 6 (B) 7 (C) 8 (D) 9 (E) 10

09b

Problem 19

A particular 12-hour digital clock displays the hour and minute of a day. Unfortunately, whenever it is supposed to display a 1, it mistakenly displays a 9. For example, when it is 1:16 PM the clock incorrectly shows 9:96 PM. What fraction of the day will the clock show the correct time?

- (A) $\frac{1}{2}$ (B) $\frac{5}{8}$ (C) $\frac{3}{4}$ (D) $\frac{5}{6}$ (E) $\frac{9}{10}$

10a

Problem 21

The polynomial $x^3 - ax^2 + bx - 2010$ has three positive integer roots. What is the smallest possible value of a ?

- (A) 78 (B) 88 (C) 98 (D) 108 (E) 118

Problem 23

Each of 2010 boxes in a line contains a single red marble, and for $1 \leq k \leq 2010$, the box in the k th position also contains k white marbles. Isabella begins at the first box and successively draws a single marble at random from each box, in order. She stops when she first draws a red marble. Let $P(n)$ be the probability that Isabella stops after drawing exactly n marbles. What is the smallest value of n for which $P(n) < \frac{1}{2010}$?

- (A) 45 (B) 63 (C) 64 (D) 201 (E) 1005

10b

Problem 17

Every high school in the city of Euclid sent a team of 3 students to a math contest. Each participant in the contest received a different score. Andrea's score was the median among all students, and hers was the highest score on her team. Andrea's teammates Beth and Carla placed 37th and 64th, respectively. How many schools are in the city?

- (A) 22 (B) 23 (C) 24 (D) 25 (E) 26

Problem 24

A high school basketball game between the Raiders and Wildcats was tied at the end of the first quarter. The number of points scored by the Raiders in each of the four quarters formed an increasing geometric sequence, and the number of points scored by the Wildcats in each of the four quarters formed an increasing arithmetic sequence. At the end of the fourth quarter, the Raiders had won by one point. Neither team scored more than 100 points. What was the total number of points scored by the two teams in the first half?

- (A) 30 (B) 31 (C) 32 (D) 33 (E) 34

Problem 25

Let $a > 0$, and let $P(x)$ be a polynomial with integer coefficients such that

$$\begin{aligned} P(1) = P(3) = P(5) = P(7) &= a, \text{ and} \\ P(2) = P(4) = P(6) = P(8) &= -a. \end{aligned}$$

What is the smallest possible value of a ?

- (A) 105 (B) 315 (C) 945 (D) 7! (E) 8!

11a

Problem 16

Which of the following is equal to $\sqrt{9 - 6\sqrt{2}} + \sqrt{9 + 6\sqrt{2}}$?

- (A) $3\sqrt{2}$ (B) $2\sqrt{6}$ (C) $\frac{7\sqrt{2}}{2}$ (D) $3\sqrt{3}$ (E) 6

Problem 17

In the eight-term sequence A, B, C, D, E, F, G, H , the value of C is 5 and the sum of any three consecutive terms is 30. What is $A + H$?

- (A) 17 (B) 18 (C) 25 (D) 26 (E) 43

Problem 19

In 1991 the population of a town was a perfect square. Ten years later, after an increase of 150 people, the population was 9 more than a perfect square. Now, in 2011, with an increase of another 150 people, the population is once again a perfect square. Which of the following is closest to the percent growth of the town's population during this twenty-year period?

- (A) 42 (B) 47 (C) 52 (D) 57 (E) 62

Problem 25

Let R be a square region and $n \geq 4$ an integer. A point X in the interior of R is called n -ray partitional if there are n rays emanating from X that divide R into n triangles of equal area. How many points are 100-ray partitional but not 60-ray partitional?

- (A) 1500 (B) 1560 (C) 2320 (D) 2480 (E) 2500

11b

Problem 19

What is the product of all the roots of the equation

$$\sqrt{5|x| + 8} = \sqrt{x^2 - 16}.$$

- (A) -64 (B) -24 (C) -9 (D) 24 (E) 576

Problem 21

Brian writes down four integers $w > x > y > z$ whose sum is 44. The pairwise positive differences of these numbers are 1, 3, 4, 5, 6, and 9. What is the sum of the possible values for w ?

- (A) 16 (B) 31 (C) 48 (D) 62 (E) 93

Problem 24

A lattice point in an xy -coordinate system is any point (x, y) where both x and y are integers. The graph of $y = mx + 2$ passes through no lattice point with $0 < x \leq 100$ for all m such that $1/2 < m < a$. What is the maximum possible value of a ?

- (A) $\frac{51}{101}$ (B) $\frac{50}{99}$ (C) $\frac{51}{100}$ (D) $\frac{52}{101}$ (E) $\frac{13}{25}$

12a

Problem 16

Three runners start running simultaneously from the same point on a 500-meter circular track. They each run clockwise around the course maintaining constant speeds of 4.4, 4.8, and 5.0 meters per second. The runners stop once they are all together again somewhere on the circular course. How many seconds do the runners run?

- (A) 1,000 (B) 1,250 (C) 2,500 (D) 5,000 (E) 10,000

Problem 19

Paula the painter and her two helpers each paint at constant, but different, rates. They always start at 8:00 AM, and all three always take the same amount of time to eat lunch. On Monday the three of them painted 50% of a house, quitting at 4:00 PM. On Tuesday, when Paula wasn't there, the two helpers painted only 24% of the house and quit at 2:12 PM. On Wednesday Paula worked by herself and finished the house by working until 7:12 P.M. How long, in minutes, was each day's lunch break?

- (A) 30 (B) 36 (C) 42 (D) 48 (E) 60

Problem 24

Let a , b , and c be positive integers with $a \geq b \geq c$ such that

$$\begin{aligned} a^2 - b^2 - c^2 + ab &= 2011 \text{ and} \\ a^2 + 3b^2 + 3c^2 - 3ab - 2ac - 2bc &= -1997. \end{aligned}$$

What is a ?

- (A) 249 (B) 250 (C) 251 (D) 252 (E) 253

12b

13a

Problem 18

Let points $A = (0, 0)$, $B = (1, 2)$, $C = (3, 3)$, and $D = (4, 0)$. Quadrilateral $ABCD$ is cut into equal area pieces by a line passing through A . This line intersects $\overline{CD}$ at point $\left(\frac{p}{q}, \frac{r}{s}\right)$, where these fractions are in lowest terms. What is $p + q + r + s$?

- (A) 54 (B) 58 (C) 62 (D) 70 (E) 75

13b

Problem 19

The real numbers c, b, a form an arithmetic sequence with $a \geq b \geq c \geq 0$. The quadratic $ax^2 + bx + c$ has exactly one root. What is this root?

- (A) $-7 - 4\sqrt{3}$ (B) $-2 - \sqrt{3}$ (C) -1 (D) $-2 + \sqrt{3}$ (E) $-7 + 4\sqrt{3}$

14a

Problem 21

Positive integers a and b are such that the graphs of $y = ax + 5$ and $y = 3x + b$ intersect the x -axis at the same point. What is the sum of all possible x -coordinates of these points of intersection?

- (A) -20 (B) -18 (C) -15 (D) -12 (E) -8

14b

Problem 20

For how many integers x is the number $x^4 - 51x^2 + 50$ negative?

- (A) 8 (B) 10 (C) 12 (D) 14 (E) 16

15a

Problem 16

If $y + 4 = (x - 2)^2$, $x + 4 = (y - 2)^2$, and $x \neq y$, what is the value of $x^2 + y^2$?

- (A) 10 (B) 15 (C) 20 (D) 25 (E) 30

Problem 17

A line that passes through the origin intersects both the line $x = 1$ and the line $y = 1 + \frac{\sqrt{3}}{3}x$. The three lines create an equilateral triangle. What is the perimeter of the triangle?

- (A) $2\sqrt{6}$ (B) $2 + 2\sqrt{3}$ (C) 6 (D) $3 + 2\sqrt{3}$ (E) $6 + \frac{\sqrt{3}}{3}$

Problem 23

The zeroes of the function $f(x) = x^2 - ax + 2a$ are integers. What is the sum of the possible values of a ?

- (A) 7 (B) 8 (C) 16 (D) 17 (E) 18

15b

Problem 24

Aaron the ant walks on the coordinate plane according to the following rules. He starts at the origin $p_0 = (0, 0)$ facing to the east and walks one unit, arriving at $p_1 = (1, 0)$. For $n = 1, 2, 3, \dots$, right after arriving at the point p_n , if Aaron can turn 90° left and walk one unit to an unvisited point p_{n+1} , he does that. Otherwise, he walks one unit straight ahead to reach p_{n+1} . Thus the sequence of points continues $p_2 = (1, 1)$, $p_3 = (0, 1)$, $p_4 = (-1, 1)$, $p_5 = (-1, 0)$, and so on in a counterclockwise spiral pattern. What is p_{2015} ?

- (A) $(-22, -13)$ (B) $(-13, -22)$ (C) $(-13, 22)$ (D) $(13, -22)$ (E) $(22, -13)$

16a

Problem 23

A binary operation $\diamond$ has the properties that $a \diamond (b \diamond c) = (a \diamond b) \cdot c$ and that $a \diamond a = 1$ for all nonzero real numbers a , b , and c . (Here $\cdot$ represents multiplication). The solution to the equation $2016 \diamond (6 \diamond x) = 100$ can be written as $\frac{p}{q}$, where p and q are relatively prime positive integers. What is $p + q$?

- (A) 109 (B) 201 (C) 301 (D) 3049 (E) 33,601

16b

Problem 16

The sum of an infinite geometric series is a positive number S , and the second term in the series is $\frac{1}{S}$. What is the smallest possible value of S ?

- (A) $\frac{1+\sqrt{5}}{2}$ (B) 2 (C) $\sqrt{5}$ (D) 3 (E) 4

Problem 20

A dilation of the plane—that is, a size transformation with a positive scale factor—sends the circle of radius 2 centered at $A(2, 2)$ to the circle of radius 3 centered at $A'(5, 6)$. What distance does the origin $O(0, 0)$ move under this transformation?

- (A) 0 (B) 3 (C) $\sqrt{13}$ (D) 4 (E) 5

Problem 21

What is the area of the region enclosed by the graph of the equation $x^2 + y^2 = |x| + |y|$?

- (A) $\pi + \sqrt{2}$ (B) $\pi + 2$ (C) $\pi + 2\sqrt{2}$ (D) $2\pi + \sqrt{2}$ (E) $2\pi + 2\sqrt{2}$

Problem 25

Let $f(x) = \sum_{k=2}^{10} ([kx] - k[x])$, where $[r]$ denotes the greatest integer less than or equal to r . How many distinct values does $f(x)$ assume for $x \geq 0$?

- (A) 32 (B) 36 (C) 45 (D) 46 (E) infinitely many

17a

Problem 17

Distinct points P, Q, R, S lie on the circle $x^2 + y^2 = 25$ and have integer coordinates. The distances PQ and RS are irrational numbers. What is the greatest possible value of the ratio $\frac{PQ}{RS}$?

- (A) 3 (B) 5 (C) $3\sqrt{5}$ (D) 7 (E) $5\sqrt{2}$

Problem 20

Let $S(n)$ equal the sum of the digits of positive integer n . For example, $S(1507) = 13$. For a particular positive integer n , $S(n) = 1274$. Which of the following could be the value of $S(n+1)$?

- (A) 1 (B) 3 (C) 12 (D) 1239 (E) 1265

Problem 24

For certain real numbers a, b , and c , the polynomial

$$g(x) = x^3 + ax^2 + x + 10$$

has three distinct roots, and each root of $g(x)$ is also a root of the polynomial

$$f(x) = x^4 + x^3 + bx^2 + 100x + c.$$

What is $f(1)$?

- (A) -9009 (B) -8008 (C) -7007 (D) -6006 (E) -5005

17b

18a

Problem 21

Which of the following describes the set of values of a for which the curves $x^2 + y^2 = a^2$ and $y = x^2 - a$ in the real xy -plane intersect at exactly 3 points?

- (A) $a = \frac{1}{4}$ (B) $\frac{1}{4} < a < \frac{1}{2}$ (C) $a > \frac{1}{4}$ (D) $a = \frac{1}{2}$ (E) $a > \frac{1}{2}$

18b

Problem 20

A function f is defined recursively by $f(1) = f(2) = 1$ and

$$f(n) = f(n-1) - f(n-2) + n$$

for all integers $n \geq 3$. What is $f(2018)$?

- (A) 2016 (B) 2017 (C) 2018 (D) 2019 (E) 2020

四、数论

09b

Problem 21

What is the remainder when $3^0 + 3^1 + 3^2 + \cdots + 3^{2009}$ is divided by 8?

- (A) 0 (B) 1 (C) 2 (D) 4 (E) 6

10a

Problem 24

The number obtained from the last two nonzero digits of $90!$ is equal to n . What is n ?

- (A) 12 (B) 32 (C) 48 (D) 52 (E) 68

Problem 25

Jim starts with a positive integer n and creates a sequence of numbers. Each successive number is obtained by subtracting the largest possible integer square less than or equal to the current number until zero is reached. For example, if Jim starts with $n = 55$, then his sequence contains 5 numbers:

$$\begin{array}{rclcl} & & & & 55 \\ 55 & - & 7^2 & = & 6 \\ 6 & - & 2^2 & = & 2 \\ 2 & - & 1^2 & = & 1 \\ 1 & - & 1^2 & = & 0 \end{array}$$

Let N be the smallest number for which Jim's sequence has 8 numbers. What is the units digit of N ?

- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

10b

Problem 18

Positive integers a , b , and c are randomly and independently selected with replacement from the set $\{1, 2, 3, \dots, 2010\}$. What is the probability that $abc + ab + a$ is divisible by 3?

- (A) $\frac{1}{3}$ (B) $\frac{29}{81}$ (C) $\frac{31}{81}$ (D) $\frac{11}{27}$ (E) $\frac{13}{27}$

11a

11b

Problem 23

What is the hundreds digit of 2011^{2011} ?

- (A) 1 (B) 4 (C) 5 (D) 6 (E) 9

12a

Problem 17

Let a and b be relatively prime integers with $a > b > 0$ and $\frac{a^3 - b^3}{(a - b)^3} = \frac{73}{3}$. What is $a - b$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 22

The sum of the first m positive odd integers is 212 more than the sum of the first n positive even integers. What is the sum of all possible values of n ?

- (A) 255 (B) 256 (C) 257 (D) 258 (E) 259

12b

Problem 20

Bernardo and Silvia play the following game. An integer between 0 and 999, inclusive, is selected and given to Bernardo. Whenever Bernardo receives a number, he doubles it and passes the result to Silvia. Whenever Silvia receives a number, she adds 50 to it and passes the result to Bernardo. The winner is the last person who produces a number less than 1000. Let N be the smallest initial number that results in a win for Bernardo. What is the sum of the digits of N ?

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Problem 22

Let $(a_1, a_2, \dots, a_{10})$ be a list of the first 10 positive integers such that for each $2 \leq i \leq 10$ either $a_i + 1$ or $a_i - 1$ or both appear somewhere before a_i in the list. How many such lists are there?

- (A) 120 (B) 512 (C) 1024 (D) 181,440 (E) 362,880

13a

Problem 17

Daphne is visited periodically by her three best friends: Alice, Beatrix, and Claire. Alice visits every third day, Beatrix visits every fourth day, and Claire visits every fifth day. All three friends visited Daphne yesterday. How many days of the next 365-day period will exactly two friends visit her?

- (A) 48 (B) 54 (C) 60 (D) 66 (E) 72

Problem 19

In base 10, the number 2013 ends in the digit 3. In base 9, on the other hand, the same number is written as $(2676)_9$ and ends in the digit 6. For how many positive integers b does the base- b -representation of 2013 end in the digit 3?

- (A) 6 (B) 9 (C) 13 (D) 16 (E) 18

Problem 21

A group of 12 pirates agree to divide a treasure chest of gold coins among themselves as follows. The k^{th} pirate to take a share takes $\frac{k}{12}$ of the coins that remain in the chest. The number of coins initially in the chest is the smallest number for which this arrangement will allow each pirate to receive a positive whole number of coins. How many coins does the 12th pirate receive?

- (A) 720 (B) 1296 (C) 1728 (D) 1925 (E) 3850

Problem 25

All 20 diagonals are drawn in a regular octagon. At how many distinct points in the interior of the octagon (not on the boundary) do two or more diagonals intersect?

- (A) 49 (B) 65 (C) 70 (D) 96 (E) 128

13b

Problem 18

The number 2013 has the property that its units digit is the sum of its other digits, that is $2 + 0 + 1 = 3$. How many integers less than 2013 but greater than 1000 share this property?

- (A) 33 (B) 34 (C) 45 (D) 46 (E) 58

Problem 20

The number 2013 is expressed in the form

$$2013 = \frac{a_1!a_2!\cdots a_m!}{b_1!b_2!\cdots b_n!},$$

where $a_1 \geq a_2 \geq \cdots \geq a_m$ and $b_1 \geq b_2 \geq \cdots \geq b_n$ are positive integers and $a_1 + b_1$ is as small as possible. What is $|a_1 - b_1|$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 21

Two non-decreasing sequences of nonnegative integers have different first terms. Each sequence has the property that each term beginning with the third is the sum of the previous two terms, and the seventh term of each sequence is N . What is the smallest possible value of N ?

- (A) 55 (B) 89 (C) 104 (D) 144 (E) 273

Problem 22

The regular octagon $ABCDEFGH$ has its center at J . Each of the vertices and the center are to be associated with one of the digits 1 through 9, with each digit used once, in such a way that the sums of the numbers on the lines AJE , BJF , CJG , and DJH are all equal. In how many ways can this be done?

- (A) 384 (B) 576 (C) 1152 (D) 1680 (E) 3456

Problem 24

A positive integer n is nice if there is a positive integer m with exactly four positive divisors (including 1 and m) such that the sum of the four divisors is equal to n . How many numbers in the set $\{2010, 2011, 2012, \dots, 2019\}$ are nice?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 25

Bernardo chooses a three-digit positive integer N and writes both its base-5 and base-6 representations on a blackboard. Later LeRoy sees the two numbers Bernardo has written. Treating the two numbers as base-10 integers, he adds them to obtain an integer S . For example, if $N = 749$, Bernardo writes the numbers 10,444 and 3,245, and LeRoy obtains the sum $S = 13,689$. For how many choices of N are the two rightmost digits of S , in order, the same as those of $2N$?

- (A) 5 (B) 10 (C) 15 (D) 20 (E) 25

14a

Problem 20

The product $(8)(888\dots 8)$, where the second factor has k digits, is an integer whose digits have a sum of 1000. What is k ?

- (A) 901 (B) 911 (C) 919 (D) 991 (E) 999

Problem 24

A sequence of natural numbers is constructed by listing the first 4, then skipping one, listing the next 5, skipping 2, listing 6, skipping 3, and, on the n th iteration, listing $n+3$ and skipping n . The sequence begins 1, 2, 3, 4, 6, 7, 8, 9, 10, 13. What is the 500,000th number in the sequence?

- (A) 996,506 (B) 996,507 (C) 996,508 (D) 996,509 (E) 996,510

Problem 25

The number 5^{867} is between 2^{2013} and 2^{2014} . How many pairs of integers (m, n) are there such that $1 \leq m \leq 2012$ and

$$5^n < 2^m < 2^{m+2} < 5^{n+1}?$$

- (A) 278 (B) 279 (C) 280 (D) 281 (E) 282

14b

Problem 17

What is the greatest power of 2 that is a factor of $10^{1002} - 4^{501}$?

- (A) 2^{1002} (B) 2^{1003} (C) 2^{1004} (D) 2^{1005} (E) 2^{1006}

Problem 18

A list of 11 positive integers has a mean of 10, a median of 9, and a unique mode of 8. What is the largest possible value of an integer in the list?

- (A) 24 (B) 30 (C) 31 (D) 33 (E) 35

15a

Problem 18

Hexadecimal (base-16) numbers are written using numeric digits 0 through 9 as well as the letters A through F to represent 10 through 15. Among the first 1000 positive integers, there are n whose hexadecimal representation contains only numeric digits. What is the sum of the digits of n ?

- (A) 17 (B) 18 (C) 19 (D) 20 (E) 21

Problem 24

For some positive integers p , there is a quadrilateral $ABCD$ with positive integer side lengths, perimeter p , right angles at B and C , $AB = 2$, and $CD = AD$. How many different values of $p < 2015$ are possible?

- (A) 30 (B) 31 (C) 61 (D) 62 (E) 63

15b

Problem 23

Let n be a positive integer greater than 4 such that the decimal representation of $n!$ ends in k zeros and the decimal representation of $(2n)!$ ends in $3k$ zeros. Let s denote the sum of the four least possible values of n . What is the sum of the digits of s ?

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

16a

Problem 22

For some positive integer n , the number $110n^3$ has 110 positive integer divisors, including 1 and the number $110n^3$. How many positive integer divisors does the number $81n^4$ have?

- (A) 110 (B) 191 (C) 261 (D) 325 (E) 425

Problem 25

How many ordered triples (x, y, z) of positive integers satisfy $\text{lcm}(x, y) = 72$, $\text{lcm}(x, z) = 600$ and $\text{lcm}(y, z) = 900$?

- (A) 15 (B) 16 (C) 24 (D) 27 (E) 64

16b

Problem 24

How many four-digit integers $abcd$, with $a \neq 0$, have the property that the three two-digit integers $ab < bc < cd$ form an increasing arithmetic sequence? One such number is 4692, where $a = 4$, $b = 6$, $c = 9$, and $d = 2$.

- (A) 9 (B) 15 (C) 16 (D) 17 (E) 20

17a

Problem 16

There are 10 horses, named Horse 1, Horse 2, ..., Horse 10. They get their names from how many minutes it takes them to run one lap around a circular race track: Horse k runs one lap in exactly k minutes. At time 0 all the horses are together at the starting point on the track. The horses start running in the same direction, and they keep running around the circular track at their constant speeds. The least time $S > 0$, in minutes, at which all 10 horses will again simultaneously be at the starting point is $S = 2520$. Let $T > 0$ be the least time, in minutes, such that at least 5 of the horses are again at the starting point. What is the sum of the digits of T ?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Problem 25

How many integers between 100 and 999, inclusive, have the property that some permutation of its digits is a multiple of 11 between 100 and 999? For example, both 121 and 211 have this property.

- (A) 226 (B) 243 (C) 270 (D) 469 (E) 486

17b

Problem 16

How many of the base-ten numerals for the positive integers less than or equal to 2017 contain the digit 0?

- (A) 469 (B) 471 (C) 475 (D) 478 (E) 481

Problem 17

Call a positive integer *monotonous* if it is a one-digit number or its digits, when read from left to right, form either a strictly increasing or a strictly decreasing sequence. For example, 3, 23578, and 987620 are monotonous, but 88, 7434, and 23557 are not. How many monotonous positive integers are there?

- (A) 1024 (B) 1524 (C) 1533 (D) 1536 (E) 2048

Problem 23

Let $N = 123456789101112 \dots 4344$ be the 79-digit number that is formed by writing the integers from 1 to 44 in order, one after the other. What is the remainder when N is divided by 45?

- (A) 1 (B) 4 (C) 9 (D) 18 (E) 44

Problem 25

Last year Isabella took 7 math tests and received 7 different scores, each an integer between 91 and 100, inclusive. After each test she noticed that the average of her test scores was an integer. Her score on the seventh test was 95. What was her score on the sixth test?

- (A) 92 (B) 94 (C) 96 (D) 98 (E) 100

18a

Problem 17

Let S be a set of 6 integers taken from $\{1, 2, \dots, 12\}$ with the property that if a and b are elements of S with $a < b$, then b is not a multiple of a . What is the least possible value of an element in S ?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 7

Problem 18

How many nonnegative integers can be written in the form

$$a_7 \cdot 3^7 + a_6 \cdot 3^6 + a_5 \cdot 3^5 + a_4 \cdot 3^4 + a_3 \cdot 3^3 + a_2 \cdot 3^2 + a_1 \cdot 3^1 + a_0 \cdot 3^0,$$

where $a_i \in \{-1, 0, 1\}$ for $0 \leq i \leq 7$?

- (A) 512 (B) 729 (C) 1094 (D) 3281 (E) 59,048

Problem 22

Let a, b, c , and d be positive integers such that $\gcd(a, b) = 24$, $\gcd(b, c) = 36$, $\gcd(c, d) = 54$, and $70 < \gcd(d, a) < 100$. Which of the following must be a divisor of a ?

- (A) 5 (B) 7 (C) 11 (D) 13 (E) 17

Problem 25

For a positive integer n and nonzero digits a, b , and c , let A_n be the n -digit integer each of whose digits is equal to a ; let B_n be the n -digit integer each of whose digits is equal to b , and let C_n be the $2n$ -digit (not n -digit) integer each of whose digits is equal to c . What is the greatest possible value of $a + b + c$ for which there are at least two values of n such that $C_n - B_n = A_n^2$?

- (A) 12 (B) 14 (C) 16 (D) 18 (E) 20

18b

Problem 16

Let $a_1, a_2, \dots, a_{2018}$ be a strictly increasing sequence of positive integers such that

$$a_1 + a_2 + \dots + a_{2018} = 2018^{2018}.$$

What is the remainder when $a_1^3 + a_2^3 + \dots + a_{2018}^3$ is divided by 6?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Problem 19

Joey and Chloe and their daughter Zoe all have the same birthday. Joey is 1 year older than Chloe, and Zoe is exactly 1 year old today. Today is the first of the 9 birthdays on which Chloe's age will be an integral multiple of Zoe's age. What will be the sum of the two digits of Joey's age the next time his age is a multiple of Zoe's age?

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Problem 21

Mary chose an even 4-digit number n . She wrote down all the divisors of n in increasing order from left to right: $1, 2, \dots, \frac{n}{2}, n$. At some moment Mary wrote 323 as a divisor of n . What is the smallest possible value of the next divisor written to the right of 323?

- (A) 324 (B) 330 (C) 340 (D) 361 (E) 646

Problem 23

How many ordered pairs (a, b) of positive integers satisfy the equation

$$a \cdot b + 63 = 20 \cdot \text{lcm}(a, b) + 12 \cdot \text{gcd}(a, b),$$

where $\text{gcd}(a, b)$ denotes the greatest common divisor of a and b , and $\text{lcm}(a, b)$ denotes their least common multiple?

- (A) 0 (B) 2 (C) 4 (D) 6 (E) 8

Problem 25

Let $\lfloor x \rfloor$ denote the greatest integer less than or equal to x . How many real numbers x satisfy the equation $x^2 + 10,000\lfloor x \rfloor = 10,000x$?

- (A) 197 (B) 198 (C) 199 (D) 200 (E) 201

五、排列组合

Circle A has radius 100. Circle B has an integer radius $r < 100$ and remains internally tangent to circle A as it rolls once around the circumference of circle A . The two circles have the same points of tangency at the beginning and end of circle B 's trip. How many possible values can r have?

- (A) 4 (B) 8 (C) 9 (D) 50 (E) 90

Problem 22

Two cubical dice each have removable numbers 1 through 6. The twelve numbers on the two dice are removed, put into a bag, then drawn one at a time and randomly reattached to the faces of the cubes, one number to each face. The dice are then rolled and the numbers on the two top faces are added. What is the probability that the sum is 7?

- (A) $\frac{1}{9}$ (B) $\frac{1}{8}$ (C) $\frac{1}{6}$ (D) $\frac{2}{11}$ (E) $\frac{1}{5}$

Problem 24

Three distinct vertices of a cube are chosen at random. What is the probability that the plane determined by these three vertices contains points inside the cube?

- (A) $\frac{1}{4}$ (B) $\frac{3}{8}$ (C) $\frac{4}{7}$ (D) $\frac{5}{7}$ (E) $\frac{3}{4}$

09b

Problem 23

Rachel and Robert run on a circular track. Rachel runs counterclockwise and completes a lap every 90 seconds, and Robert runs clockwise and completes a lap every 80 seconds. Both start from the same line at the same time. At some random time between 10 minutes and 11 minutes after they begin to run, a photographer standing inside the track takes a picture that shows one-fourth of the track, centered on the starting line. What is the probability that both Rachel and Robert are in the picture?

- (A) $\frac{1}{16}$ (B) $\frac{1}{8}$ (C) $\frac{3}{16}$ (D) $\frac{1}{4}$ (E) $\frac{5}{16}$

Problem 25

Each face of a cube is given a single narrow stripe painted from the center of one edge to the center of the opposite edge. The choice of the edge pairing is made at random and independently for each face. What is the probability that there is a continuous stripe encircling the cube?

- (A) $\frac{1}{8}$ (B) $\frac{3}{16}$ (C) $\frac{1}{4}$ (D) $\frac{3}{8}$ (E) $\frac{1}{2}$

10a

Problem 18

Bernardo randomly picks 3 distinct numbers from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and arranges them in descending order to form a 3-digit number. Silvia randomly picks 3 distinct numbers from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$ and also arranges them in descending order to form a 3-digit number. What is the probability that Bernardo's number is larger than Silvia's number?

- (A) $\frac{47}{72}$ (B) $\frac{37}{56}$ (C) $\frac{2}{3}$ (D) $\frac{49}{72}$ (E) $\frac{39}{56}$

Problem 22

Eight points are chosen on a circle, and chords are drawn connecting every pair of points. No three chords intersect in a single point inside the circle. How many triangles with all three vertices in the interior of the circle are created?

- (A) 28 (B) 56 (C) 70 (D) 84 (E) 140

10b

Problem 21

A palindrome between 1000 and 10,000 is chosen at random. What is the probability that it is divisible by 7?

- (A) $\frac{1}{10}$ (B) $\frac{1}{9}$ (C) $\frac{1}{7}$ (D) $\frac{1}{6}$ (E) $\frac{1}{5}$

Problem 22

Seven distinct pieces of candy are to be distributed among three bags. The red bag and the blue bag must each receive at least one piece of candy; the white bag may remain empty. How many arrangements are possible?

- (A) 1930 (B) 1931 (C) 1932 (D) 1933 (E) 1934

Problem 23

The entries in a 3×3 array include all the digits from 1 through 9, arranged so that the entries in every row and column are in increasing order. How many such arrays are there?

- (A) 18 (B) 24 (C) 36 (D) 42 (E) 60

11a

Problem 20

Two points on the circumference of a circle of radius r are selected independently and at random. From each point a chord of length r is drawn in a clockwise direction. What is the probability that the two chords intersect?

- (A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

Problem 21

Two counterfeit coins of equal weight are mixed with 8 identical genuine coins. The weight of each of the counterfeit coins is different from the weight of each of the genuine coins. A pair of coins is selected at random without replacement from the 10 coins. A second pair is selected at random without replacement from the remaining 8 coins. The combined weight of the first pair is equal to the combined weight of the second pair. What is the probability that all 4 selected coins are genuine?

- (A) $\frac{7}{11}$ (B) $\frac{9}{13}$ (C) $\frac{11}{15}$ (D) $\frac{15}{19}$ (E) $\frac{15}{16}$

Problem 22

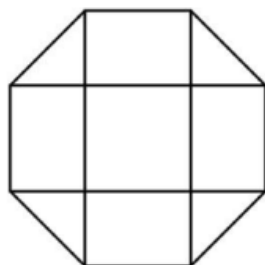
Each vertex of convex pentagon $ABCDE$ is to be assigned a color. There are 6 colors to choose from, and the ends of each diagonal must have different colors. How many different colorings are possible?

- (A) 2520 (B) 2880 (C) 3120 (D) 3250 (E) 3750

11b

Problem 16

A dart board is a regular octagon divided into regions as shown. Suppose that a dart thrown at the board is equally likely to land anywhere on the board. What is probability that the dart lands within the center square?



- (A) $\frac{\sqrt{2}-1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{2-\sqrt{2}}{2}$ (D) $\frac{\sqrt{2}}{4}$ (E) $2-\sqrt{2}$

12a

Problem 20

A 3×3 square is partitioned into 9 unit squares. Each unit square is painted either white or black with each color being equally likely, chosen independently and at random. The square is then rotated 90° clockwise about its center, and every white square in a position formerly occupied by a black square is painted black. The colors of all other squares are left unchanged. What is the probability the grid is now entirely black?

- (A) $\frac{49}{512}$ (B) $\frac{7}{64}$ (C) $\frac{121}{1024}$ (D) $\frac{81}{512}$ (E) $\frac{9}{32}$

Problem 23

Adam, Benin, Chiang, Deshawn, Esther, and Fiona have internet accounts. Some, but not all, of them are internet friends with each other, and none of them has an internet friend outside this group. Each of them has the same number of internet friends. In how many different ways can this happen?

- (A) 60 (B) 170 (C) 290 (D) 320 (E) 660

Problem 25

Real numbers x , y , and z are chosen independently and at random from the interval $[0, n]$ for some positive integer n . The probability that no two of x , y , and z are within 1 unit of each other is greater than $\frac{1}{2}$. What is the smallest possible value of n ?

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

12b

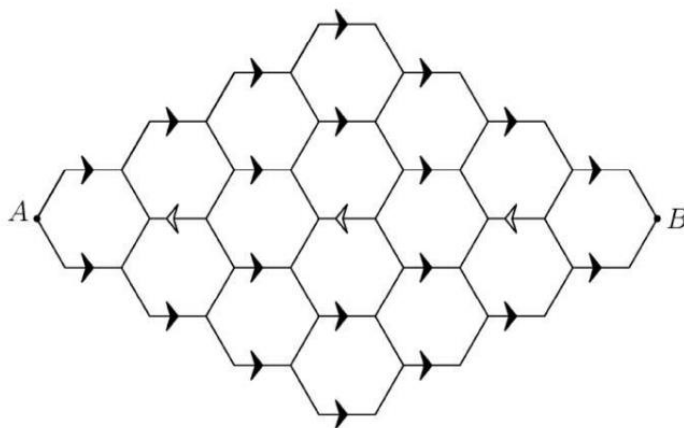
Problem 24

Amy, Beth, and Jo listen to four different songs and discuss which ones they like. No song is liked by all three. Furthermore, for each of the three pairs of the girls, there is at least one song liked by those girls but disliked by the third. In how many different ways is this possible?

- (A) 108 (B) 132 (C) 671 (D) 846 (E) 1105

Problem 25

A bug travels from A to B along the segments in the hexagonal lattice pictured below. The segments marked with an arrow can be traveled only in the direction of the arrow, and the bug never travels the same segment more than once. How many different paths are there?



- (A) 2112 (B) 2304 (C) 2368 (D) 2384 (E) 2400

13b

Problem 17

Alex has 75 red tokens and 75 blue tokens. There is a booth where Alex can give two red tokens and receive in return a silver token and a blue token, and another booth where Alex can give three blue tokens and receive in return a silver token and a red token. Alex continues to exchange tokens until no more exchanges are possible. How many silver tokens will Alex have at the end?

- (A) 62 (B) 82 (C) 83 (D) 102 (E) 103

14a

Problem 17

Three fair six-sided dice are rolled. What is the probability that the values shown on two of the dice sum to the value shown on the remaining die?

- (A) $\frac{1}{6}$ (B) $\frac{13}{72}$ (C) $\frac{7}{36}$ (D) $\frac{5}{24}$ (E) $\frac{2}{9}$

14b

Problem 16

Four fair six-sided dice are rolled. What is the probability that at least three of the four dice show the same value?

- (A) $\frac{1}{36}$ (B) $\frac{7}{72}$ (C) $\frac{1}{9}$ (D) $\frac{5}{36}$ (E) $\frac{1}{6}$

Problem 19

Two concentric circles have radii **1** and **2**. Two points on the outer circle are chosen independently and uniformly at random. What is the probability that the chord joining the two points intersects the inner circle?

- (A) $\frac{1}{6}$ (B) $\frac{1}{4}$ (C) $\frac{2-\sqrt{2}}{2}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

Problem 24

The numbers 1, 2, 3, 4, 5 are to be arranged in a circle. An arrangement is bad if it is not true that for every n from **1** to **15** one can find a subset of the numbers that appear consecutively on the circle that sum to n . Arrangements that differ only by a rotation or a reflection are considered the same. How many different bad arrangements are there?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 25

In a small pond there are eleven lily pads in a row labeled **0** through **10**. A frog is sitting on pad **1**. When the frog is on pad N , $0 < N < 10$, it will jump to pad $N-1$ with probability $\frac{N}{10}$ and to pad $N+1$ with probability $1 - \frac{N}{10}$. Each jump is independent of the previous jumps. If the frog reaches pad **0** it will be eaten by a patiently waiting snake. If the frog reaches pad **10** it will exit the pond, never to return. What is the probability that the frog will escape being eaten by the snake?

- (A) $\frac{32}{79}$ (B) $\frac{161}{384}$ (C) $\frac{63}{146}$ (D) $\frac{7}{16}$ (E) $\frac{1}{2}$

15a

Problem 22

Eight people are sitting around a circular table, each holding a fair coin. All eight people flip their coins and those who flip heads stand while those who flip tails remain seated. What is the probability that no two adjacent people will stand?

- (A) $\frac{47}{256}$ (B) $\frac{3}{16}$ (C) $\frac{49}{256}$ (D) $\frac{25}{128}$ (E) $\frac{51}{256}$

15b

Problem 16

Al, Bill, and Cal will each randomly be assigned a whole number from **1** to **10**, inclusive, with no two of them getting the same number. What is the probability that Al's number will be a whole number multiple of Bill's and Bill's number will be a whole number multiple of Cal's?

- (A) $\frac{9}{1000}$ (B) $\frac{1}{90}$ (C) $\frac{1}{80}$ (D) $\frac{1}{72}$ (E) $\frac{2}{121}$

Problem 18

Johann has **64** fair coins. He flips all the coins. Any coin that lands on tails is tossed again. Coins that land on tails on the second toss are tossed a third time. What is the expected number of coins that are now heads?

- (A) 32 (B) 40 (C) 48 (D) 56 (E) 64

Problem 21

Cozy the Cat and Dash the Dog are going up a staircase with a certain number of steps. However, instead of walking up the steps one at a time, both Cozy and Dash jump. Cozy goes two steps up with each jump (though if necessary, he will just jump the last step). Dash goes five steps up with each jump (though if necessary, he will just jump the last steps if there are fewer than 5 steps left). Suppose Dash takes 19 fewer jumps than Cozy to reach the top of the staircase. Let S denote the sum of all possible numbers of steps this staircase can have. What is the sum of the digits of S ?

- (A) 9 (B) 11 (C) 12 (D) 13 (E) 15

Problem 25

A rectangular box measures $a \times b \times c$, where a, b , and c are integers and $1 \leq a \leq b \leq c$. The volume and surface area of the box are numerically equal. How many ordered triples (a, b, c) are possible?

- (A) 4 (B) 10 (C) 12 (D) 21 (E) 26

16a

Problem 17

Let N be a positive multiple of 5. One red ball and N green balls are arranged in a line in random order. Let $P(N)$ be the probability that at least $\frac{3}{5}$ of the green balls are on the same side of the red ball. Observe that $P(5) = 1$ and that $P(N)$ approaches $\frac{4}{5}$ as N grows large. What is the sum of the digits of the least value of N such that $P(N) < \frac{321}{400}$?

- (A) 12 (B) 14 (C) 16 (D) 18 (E) 20

Problem 20

For some particular value of N , when $(a + b + c + d + 1)^N$ is expanded and like terms are combined, the resulting expression contains exactly 1001 terms that include all four variables a, b, c , and d , each to some positive power. What is N ?

- (A) 9 (B) 14 (C) 16 (D) 17 (E) 19

16b

Problem 17

All the numbers 2, 3, 4, 5, 6, 7 are assigned to the six faces of a cube, one number to each face. For each of the eight vertices of the cube, a product of three numbers is computed, where the three numbers are the numbers assigned to the three faces that include that vertex. What is the greatest possible value of the sum of these eight products?

- (A) 312 (B) 343 (C) 625 (D) 729 (E) 1680

Problem 18

In how many ways can 345 be written as the sum of an increasing sequence of two or more consecutive positive integers?

- (A) 1 (B) 3 (C) 5 (D) 6 (E) 7

Problem 22

A set of teams held a round-robin tournament in which every team played every other team exactly once. Every team won 10 games and lost 10 games; there were no ties. How many sets of three teams $\{A, B, C\}$ were there in which A beat B , B beat C , and C beat A ?

- (A) 385 (B) 665 (C) 945 (D) 1140 (E) 1330

17a

Problem 18

Amelia has a coin that lands heads with probability $\frac{1}{3}$, and Blaine has a coin that lands on heads with probability $\frac{2}{5}$. Amelia and Blaine alternately toss their coins until someone gets a head; the first one to get a head wins. All coin tosses are independent. Amelia goes first. The probability that Amelia wins is $\frac{p}{q}$, where p and q are relatively prime positive integers. What is $q - p$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 19

Alice refuses to sit next to either Bob or Carla. Derek refuses to sit next to Eric. How many ways are there for the five of them to sit in a row of 5 chairs under these conditions?

- (A) 12 (B) 16 (C) 28 (D) 32 (E) 40

Problem 23

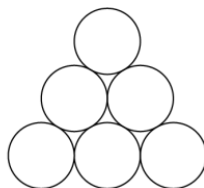
How many triangles with positive area have all their vertices at points (i, j) in the coordinate plane, where i and j are integers between 1 and 5, inclusive?

- (A) 2128 (B) 2148 (C) 2160 (D) 2200 (E) 2300

17b

Problem 18

In the figure below, 3 of the 6 disks are to be painted blue, 2 are to be painted red, and 1 is to be painted green. Two paintings that can be obtained from one another by a rotation or a reflection of the entire figure are considered the same. How many different paintings are possible?



- (A) 6 (B) 8 (C) 9 (D) 12 (E) 15

Problem 20

The number $21! = 51,090,942,171,709,440,000$ has over 60,000 positive integer divisors. One of them is chosen at random. What is the probability that it is odd?

- (A) $\frac{1}{21}$ (B) $\frac{1}{19}$ (C) $\frac{1}{18}$ (D) $\frac{1}{2}$ (E) $\frac{11}{21}$

18a

Problem 19

A number m is randomly selected from the set $\{11, 13, 15, 17, 19\}$, and a number n is randomly selected from $\{1999, 2000, 2001, \dots, 2018\}$. What is the probability that m^n has a units digit of 1?

- (A) $\frac{1}{5}$ (B) $\frac{1}{4}$ (C) $\frac{3}{10}$ (D) $\frac{7}{20}$ (E) $\frac{2}{5}$

18b

Problem 22

Real numbers x and y are chosen independently and uniformly at random from the interval $[0, 1]$. Which of the following numbers is closest to the probability that x , y , and 1 are the side lengths of an obtuse triangle?

- (A) 0.21 (B) 0.25 (C) 0.29 (D) 0.50 (E) 0.79